Ahmedabad University

ECE 501: Digital Image Processing

Week 9 Progress Report

Group 6

Vehicle Orientation Estimation Using Conventional Image Processing

Submitted to faculty: Prof. Mehul Raval

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GitHub Repository (DRASHTI-HaOBB):

https://github.com/ERYAGNIK003/DRASHTI-Dataset

Dataset Source:

DRASHTI-HaOBB – Heading-angle Oriented Bounding Box Dataset

https://zenodo.org/records/18278989

1. Work Done This Week

In Week 9, we focused on completing the project by consolidating the results obtained from the previous experiments and performing the final evaluation of the implemented vehicle orientation estimation methods.

The main objective was to bring all the experimental results together and evaluate the performance of the different approaches using the same 100 vehicle observations.

The following tasks were completed during this week:

1. Loaded the previously generated results from Weeks 3, 6, 7, and 8, including the vehicle crop manifest and the corresponding ground-truth annotations.
2. Combined the predictions from PCA, Original Hough Transform, Contour-Based Estimation, and Modified Hough Transform into a single dataset using Vehicle_ID as the common identifier.
3. Integrated the directional heading estimates generated using the Modified Hough-based axis estimation and silhouette asymmetry approach.
4. Evaluated the directional heading predictions using circular angular error and calculated the overall performance measures, including mean, median, maximum error, and standard deviation.
5. Compared the four undirected orientation estimation methods using prediction availability, Mean Absolute Error (MAE), median error, maximum error, and the number of high-error predictions.
6. Performed a class-wise comparison to examine the performance of the orientation estimation methods across different vehicle categories.
7. Identified the 15 vehicle observations with the highest directional heading errors and examined cases where the estimated heading was almost opposite to the ground-truth direction.
8. Analysed the availability of directional predictions separately from their angular accuracy to identify any failed predictions.
9. Generated graphs to visualize the directional error distribution, overall method comparison, class-wise performance, ground-truth versus estimated directional headings, the ten highest directional heading errors, and the vehicle class distribution among these high-error observations
10. Generated visual comparisons for all 100 vehicle observations in five batches of 20 images each.
11. Saved the final prediction tables, performance summaries, error records, graphs, and vehicle visualizations in the Week9_Results directory.

The overall Week 9 workflow was:

Loading Previous Results → Consolidating Predictions for 100 Vehicles → Directional Heading Evaluation → Comparison of Orientation Estimation Methods → Class-wise Analysis → High-error and Failed Prediction Analysis → Graphical Visualization → Saving Final Results

This completed the final numerical and visual evaluation of the project.

2. Methodology

2.1 Loading and Consolidating Previous Results

The final evaluation was performed using the existing results from the previous project stages.

The Week 3 results contained the ground-truth orientation values and predictions from PCA, Original Hough Transform, and Contour-Based Estimation. The Modified Hough results from Week 7 and the directional heading predictions from Week 8 were also included.

The Week 6 vehicle crop manifest was used to retrieve the corresponding vehicle images for the final visualizations.

All the datasets were merged using the Vehicle_ID column. This ensured that the predictions from different methods corresponded to the same vehicle observations.

The original 100 vehicle observations were retained throughout the analysis. Missing predictions were represented as NaN instead of removing the corresponding vehicles from the consolidated dataset.

The final dataset contained the estimated orientations, estimated directional headings, ground-truth values, and their corresponding angular errors.

This consolidated dataset was used as the basis for the remaining comparisons and analyses.

2.2 Comparative Evaluation of Orientation Estimation Methods

The four implemented methods were compared using the undirected orientation convention of 0°–90°.

The methods included:

- Principal Component Analysis (PCA).
- Original Hough Transform.
- Contour-Based Orientation Estimation.
- Modified Hough Transform.

All four methods were evaluated using the same 100 vehicle observations and the corresponding ground-truth orientation annotations.

The following performance measures were considered:

- Total number of vehicle observations.
- Number of valid and failed predictions.
- Prediction coverage.
- Mean Absolute Error (MAE).
- Median angular error.
- Maximum angular error.
- Standard deviation of angular error.
- Number of predictions with errors greater than or equal to 30°.

This comparison helped determine the relative performance of the methods and identify the approach that produced the lowest overall orientation estimation error.

2.3 Final Directional Heading Evaluation

The directional heading predictions generated in Week 8 were included in the final evaluation.

The Modified Hough Transform was used to estimate the longitudinal axis of each vehicle, while silhouette asymmetry analysis was used to select one of the two possible directions along that axis.

The resulting headings were represented in the range of 0°–359° and compared with the ground-truth directional headings.

Since directional headings are circular measurements, the error was calculated using the shortest angular difference:

$$E = min(|H_{est} - H_{GT}|, 360^{\circ} - |H_{est} - H_{GT}|)$$

where,

- H_{est} represents the estimated directional heading.
- H_{GT} represents the ground-truth directional heading.
- E represents the circular angular error.

The evaluation included mean, median, maximum, and standard deviation of angular errors, along with the number of successful and failed predictions.

The prediction coverage was calculated separately from angular accuracy so that the generation of a prediction was not treated as an indication of its correctness.

2.4 Class-wise and High-error Analysis

The four undirected orientation estimation methods were also compared across the different vehicle categories.

The mean angular error, median angular error, maximum angular error, and number of valid predictions were calculated separately for each category.

In addition, the directional heading predictions were sorted according to their angular errors to identify the 15 observations with the highest errors.

Particular attention was given to errors greater than or equal to 150°, as these cases could indicate that the estimated direction was nearly opposite to the actual vehicle direction.

The vehicle categories among the ten highest-error observations were also examined to understand which categories appeared most frequently in these cases.

2.5 Graphical Visualization and Final Result Saving

Graphs were generated to represent the final evaluation results and make the comparison easier to understand.

The visualizations included:

1. Histogram of directional heading errors.
2. Overall comparison of the four orientation estimation methods.
3. Class-wise comparison of the four orientation estimation methods.
4. Scatter plot comparing ground-truth and estimated directional headings.
5. Graph showing the ten vehicles with the highest directional heading errors.
6. Pie chart showing the vehicle class distribution among the ten highest-error observations.

Additionally, visual comparisons were generated for all 100 vehicle observations.

For every vehicle crop:

- Green arrow represented the estimated directional heading.
- Red arrow represented the ground-truth directional heading.
- Vehicle ID, ground-truth heading, estimated heading, and angular error were displayed.

The visualizations were generated in five batches of 20 images each.


00XD_3rEO_4KVW_vehicle_1
GT: 49.0° | Est: 49.3°
Error: 0.3°


2GuW_AKSD_y8m2_vehicle_1
GT: 182.4° | Est: 9.4°
Error: 173.0°


2h07_opct_WJn8_vehicle_1
GT: 168.4° | Est: 169.4°
Error: 1.1°


3J2E_aL5U_TE2G_vehicle_1
GT: 4.7° | Est: 181.1°
Error: 176.4°

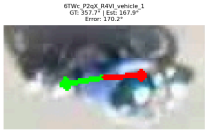

3LzM_y7ht_Sd8z_vehicle_1
GT: 24.9° | Est: 29.3°
Error: 4.4°


4AYY_W6bt_vwl6_vehicle_1
GT: 180.0° | Est: 175.0°
Error: 5.0°

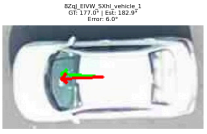

5N3r_JDF_1pzt_vehicle_1
GT: 175.1° | Est: 173.5°
Error: 1.7°


5QHK_2GHz_wRDq_vehicle_1
GT: 11.0° | Est: 133.6°
Error: 122.6°


6JBU_IC9J_Lszd_vehicle_1
GT: 191.0° | Est: 187.0°
Error: 4.0°


6TWc_P2qX_R4Vl_vehicle_1
GT: 357.7° | Est: 167.9°
Error: 170.2°

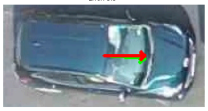

6fBI_ZIbw_EnVh_vehicle_1
GT: 60.1° | Est: 53.2°
Error: 6.9°


8ZqJ_EtVW_5Xhl_vehicle_1
GT: 177.0° | Est: 182.9°
Error: 6.0°


8eh6_NPoT_oEI1_vehicle_1
GT: 41.6° | Est: 68.5°
Error: 26.9°


95Ad_tJKX_KWb9_vehicle_1
GT: 184.7° | Est: 3.2°
Error: 178.5°

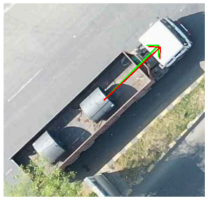

BAsS_G7XM_JuST_vehicle_1
GT: 177.6° | Est: 182.2°
Error: 4.7°


Bom3_kxhC_IAe3_vehicle_1
GT: 1.3° | Est: 5.2°
Error: 3.8°


C2g2_vYbb_ihYH_vehicle_1
GT: 239.6° | Est: 51.7°
Error: 172.1°


DF3T_rTNG_KSCM_vehicle_1
GT: 173.6° | Est: 351.8°
Error: 178.2°


G16w_936H_wBhO_vehicle_1
GT: 168.4° | Est: 168.8°
Error: 0.4°


GCwW_rFZ7_KiPJ_vehicle_1
GT: 317.0° | Est: 315.3°
Error: 1.7°


GpeL_iRkt_Hpl7_vehicle_1
GT: 360.0° | Est: 175.1°
Error: 175.1°


HLsx_XKFI_4Azz_vehicle_1
GT: 178.8° | Est: 352.2°
Error: 173.4°


I7ui_Ix5W_O2DS_vehicle_1
GT: 268.0° | Est: 270.0°
Error: 2.0°


Iwbd_XCYw_iwrB_vehicle_1
GT: 214.7° | Est: 70.5°
Error: 144.2°


JZv9_ozgt_YMRX_vehicle_1
GT: 180.7° | Est: 173.9°
Error: 6.1°


KBtE_E7BV_E33r_vehicle_1
GT: 115.2° | Est: 128.9°
Error: 13.7°


KNvT_zacV_LpYg_vehicle_1
GT: 283.2° | Est: 111.1°
Error: 172.1°


Kp67_1UBu_Rx2q_vehicle_1
GT: 330.6° | Est: 138.2°
Error: 167.6°


LGTf_z88f_Jed0_vehicle_1
GT: 274.0° | Est: 271.9°
Error: 2.1°


LOKr_rvnD_oaek_vehicle_1
GT: 60.1° | Est: 55.1°
Error: 5.0°


MOL9_poaA_17Ab_vehicle_1
GT: 180.0° | Est: 181.9°
Error: 1.9°


McY4_xL3v_IDUJ_vehicle_1
GT: 358.2° | Est: 355.0°
Error: 3.2°


NwkF_XAS3_2yXy_vehicle_1
GT: 357.7° | Est: 171.5°
Error: 173.8°


ON14_gohO_FI9V_vehicle_1
GT: 183.7° | Est: 5.7°
Error: 178.0°


OfAq_cdb3_0fXF_vehicle_1
GT: 221.8° | Est: 44.0°
Error: 177.8°


OhoE_5DTs_Orwz_vehicle_1
GT: 173.1° | Est: 172.1°
Error: 1.0°

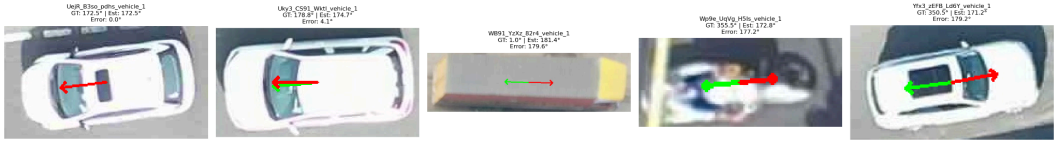

OyGP_UDOJ_hIJP_vehicle_1
GT: 86.0° | Est: 266.1°
Error: 179.9°

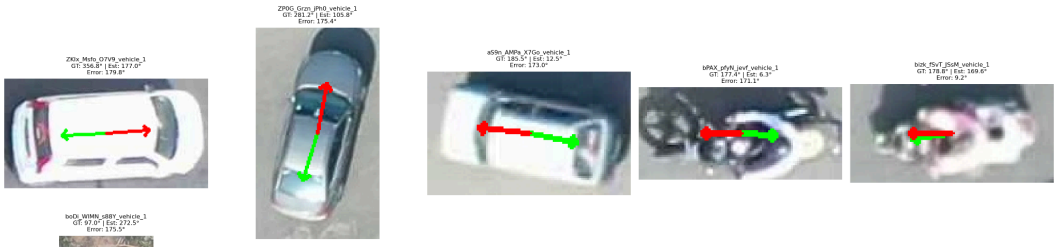

Ozpo_GT82_09Gy_vehicle_1
GT: 189.5° | Est: 12.3°
Error: 177.2°


PNK_43IC_DtXA_vehicle_1
GT: 326.7° | Est: 134.5°
Error: 167.8°


PJJB_h1vQ_Vxav_vehicle_1
GT: 170.6° | Est: 169.6°
Error: 0.9°


PnvMt_uL7n_KIk1_vehicle_1
GT: 359.1° | Est: 181.9°
Error: 177.1°
RXzz_utpH_J6LU_vehicle_1
GT: 156.1° | Est: 70.5°
Error: 85.5°
RyLX_MUJO_U7tv_vehicle_1
GT: 187.7° | Est: 13.2°
Error: 174.5°
SR9o_F2yn_rN3o_vehicle_1
GT: 8.5° | Est: 176.6°
Error: 168.2°
TJKp_GI8z_vafn_vehicle_1
GT: 180.0° | Est: 350.1°
Error: 170.1°


UejR_B3so_pdhs_vehicle_1
GT: 172.5° | Est: 172.5°
Error: 0.0°
Uky3_CS91_Wkt1_vehicle_1
GT: 178.8° | Est: 174.7°
Error: 4.1°
WB91_YzXz_82r4_vehicle_1
GT: 1.0° | Est: 181.4°
Error: 179.6°
WpYe_UqVg_HSlx_vehicle_1
GT: 355.5° | Est: 172.8°
Error: 177.2°
Yfx3_zEfB_Lo8Y_vehicle_1
GT: 350.5° | Est: 171.2°
Error: 179.2°


ZKbx_Mxfo_O7V8_vehicle_1
GT: 356.8° | Est: 177.0°
Error: 179.8°
ZP0G_Grzn_jPh0_vehicle_1
GT: 281.2° | Est: 105.8°
Error: 175.4°
aS9n_AMPs_X7Go_vehicle_1
GT: 185.5° | Est: 12.5°
Error: 173.0°
bPAX_pfyN_Jevf_vehicle_1
GT: 177.4° | Est: 6.3°
Error: 171.1°
bIzk_fSvT_JSsM_vehicle_1
GT: 178.8° | Est: 169.6°
Error: 9.2°

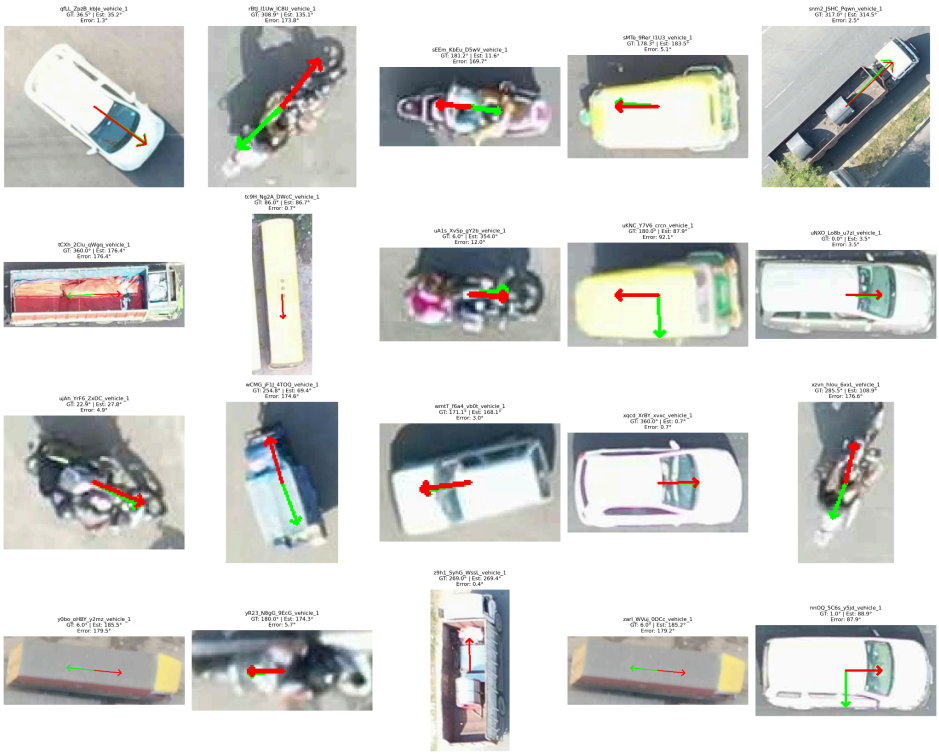

boDi_WIMN_s8BY_vehicle_1
GT: 97.0° | Est: 272.5°
Error: 175.5°
cJRl_mjsU_Z6FP_vehicle_1
GT: 0.0° | Est: 2.7°
Error: 2.7°
cLEH_vQvJ_zAUO_vehicle_1
GT: 359.0° | Est: 271.0°
Error: 88.0°
dDRH_wv1m_jsew_vehicle_1
GT: 175.9° | Est: 175.2°
Error: 0.7°
dH28_2Thg_72eh_vehicle_1
GT: 180.0° | Est: 4.1°
Error: 175.9°


dpg_d95P_P2oG_vehicle_1
GT: 174.6° | Est: 171.8°
Error: 2.7°
eBja_W8tH_Hodr_vehicle_1
GT: 33.0° | Est: 46.0°
Error: 13.0°
fhXz_KKgf_zJEr_vehicle_1
GT: 0.0° | Est: 349.9°
Error: 10.1°
gqbe_NCOI_MTLw_vehicle_1
GT: 171.3° | Est: 172.1°
Error: 0.7°
hOrt_o8Wa_4NSk_vehicle_1
GT: 179.4° | Est: 356.4°
Error: 177.0°


hnBn_2cUB_01Nx_vehicle_1
GT: 78.6° | Est: 263.6°
Error: 174.9°
haeu_D8Sq_eySG_vehicle_1
GT: 270.0° | Est: 92.0°
Error: 178.0°
ind0_yxhu_aCtu_vehicle_1
GT: 173.5° | Est: 171.0°
Error: 2.5°
jWW_Zbs6_0dnW_vehicle_1
GT: 21.7° | Est: 33.9°
Error: 12.2°
lPCV_CAQ9_X0wi_vehicle_2
GT: 140.0° | Est: 170.7°
Error: 10.7°


mCA9_HWys_Z2vx_vehicle_1
GT: 352.5° | Est: 164.6°
Error: 172.1°
nIod_CVl1_IMKE_vehicle_1
GT: 180.0° | Est: 180.9°
Error: 0.9°
nk9h_vkXM_86o7_vehicle_1
GT: 355.5° | Est: 348.9°
Error: 6.7°
oFvW_F7NA_jcZE_vehicle_1
GT: 194.3° | Est: 29.0°
Error: 167.2°
oGWr_BH90_2vh1_vehicle_1
GT: 171.7° | Est: 172.1°
Error: 0.4°


oTf5_WAoB_x73N_vehicle_1
GT: 154.3° | Est: 153.7°
Error: 0.6°
oWs3_ZpmL_rlSk_vehicle_1
GT: 80.2° | Est: 73.4°
Error: 6.7°
op2r_2VVe_JASo_vehicle_1
GT: 79.1° | Est: 90.4°
Error: 11.3°
onQ3_Jqgh_vTnF_vehicle_1
GT: 274.0° | Est: 269.4°
Error: 6.6°
qWs1_rR7t_qewx_vehicle_1
GT: 182.9° | Est: 11.5°
Error: 171.4°



All the final tables, graphs, and visualization batches were saved in the Week9_Results directory.

3. Results and Observations

3.1 Consolidated Results for the 100 Vehicle Observations

The results from the previous experiments were successfully combined into a single dataset containing 100 vehicle observations.

The prediction availability for each method is presented below.

Method	Total Vehicles	Valid Predictions	Failed Predictions
PCA	100	100	0
Original Hough	100	98	2
Contour-Based	100	100	0
Modified Hough	100	100	0
Modified Hough + Silhouette (Directional)	100	100	0

Table 1. Prediction availability across the implemented methods

The results show that all four undirected orientation estimation methods produced predictions for most of the vehicle observations. The Original Hough Transform had two unavailable predictions, whereas the other three methods generated predictions for all 100 observations.

The directional heading method also generated predictions for all 100 vehicle observations.

3.2 Comparative Performance of the Four Orientation Estimation Methods

The four undirected orientation estimation methods were compared using their corresponding angular errors.

The results are presented below.

Method	Valid Predictions	MAE	Median Error	Maximum Error	Standard Deviation	Errors ≥ 30°
PCA	100	10.74°	4.85°	81.71°	16.11°	8
Original Hough	98	12.46°	4.18°	82.39°	20.44°	11
Contour-Based	100	10.97°	5.72°	89.00°	16.44°	12
Modified Hough	100	**5.35°**	**2.02°**	84.94°	**11.12°**	**3**

Table 2. Comparative performance of the four orientation estimation methods

The Modified Hough Transform achieved the lowest Mean Absolute Error of 5.35° among the four methods. It also produced the lowest median angular error of 2.02° and the lowest standard deviation of 11.12°.

Only three predictions from the Modified Hough method had errors greater than or equal to 30°, compared to eight for PCA, eleven for Original Hough, and twelve for Contour-Based Estimation.

These results indicate that the Modified Hough Transform provided the best overall performance for undirected vehicle orientation estimation on the selected 100 observations.

However, its maximum angular error of 84.94° also shows that some individual predictions still had relatively large errors.

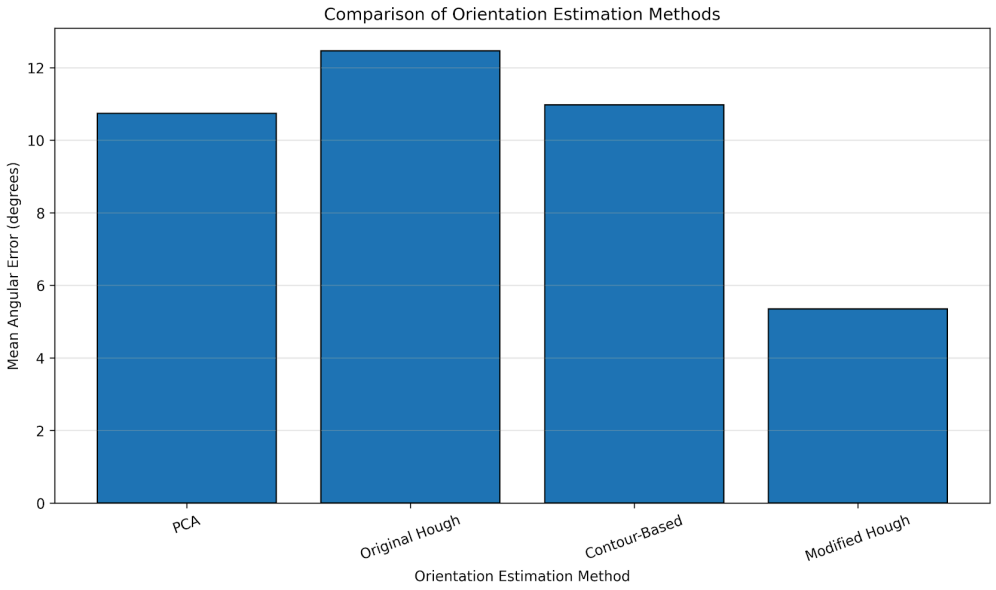


Figure 1. Comparison of Mean Absolute Error among the four orientation estimation methods.

3.3 Overall Directional Heading Estimation Results

The directional heading predictions were compared with the ground-truth headings using circular angular error.

The final performance measures are presented below.

Performance Measure	Result
Total Vehicle Observations	100
Successfully Generated Predictions	100
Failed Predictions	0
Prediction Coverage	100%
Mean Absolute Error	82.02°
Median Angular Error	13.35°
Maximum Angular Error	179.91°
Standard Deviation	82.83°
Vehicles with Errors ≥ 30°	48

Table 3. Overall performance of directional heading estimation

The results show that the directional heading estimation method successfully generated predictions for all 100 vehicle observations. However, the mean angular error was 82.02°, indicating that the estimated headings did not consistently match the ground-truth directions.

The maximum error was about 179.91° and the median error was 13.35°. The large difference suggests that there were a few observations with relatively small errors, while there were also a few predictions that were almost the opposite of the actual direction of the vehicle.

Out of the 100 observations, 48 had errors greater than or equal to 30°.

The Week 8 axis-only evaluation had also shown a mean axis error of 9.09°, a median error of 4.46°, and 94 out of 100 observations with errors of 30° or less. This comparison indicates that estimating the longitudinal axis was considerably more reliable than identifying the correct front–rear direction.

The directional heading results therefore highlight the additional difficulty introduced by resolving the 180° ambiguity using silhouette asymmetry.

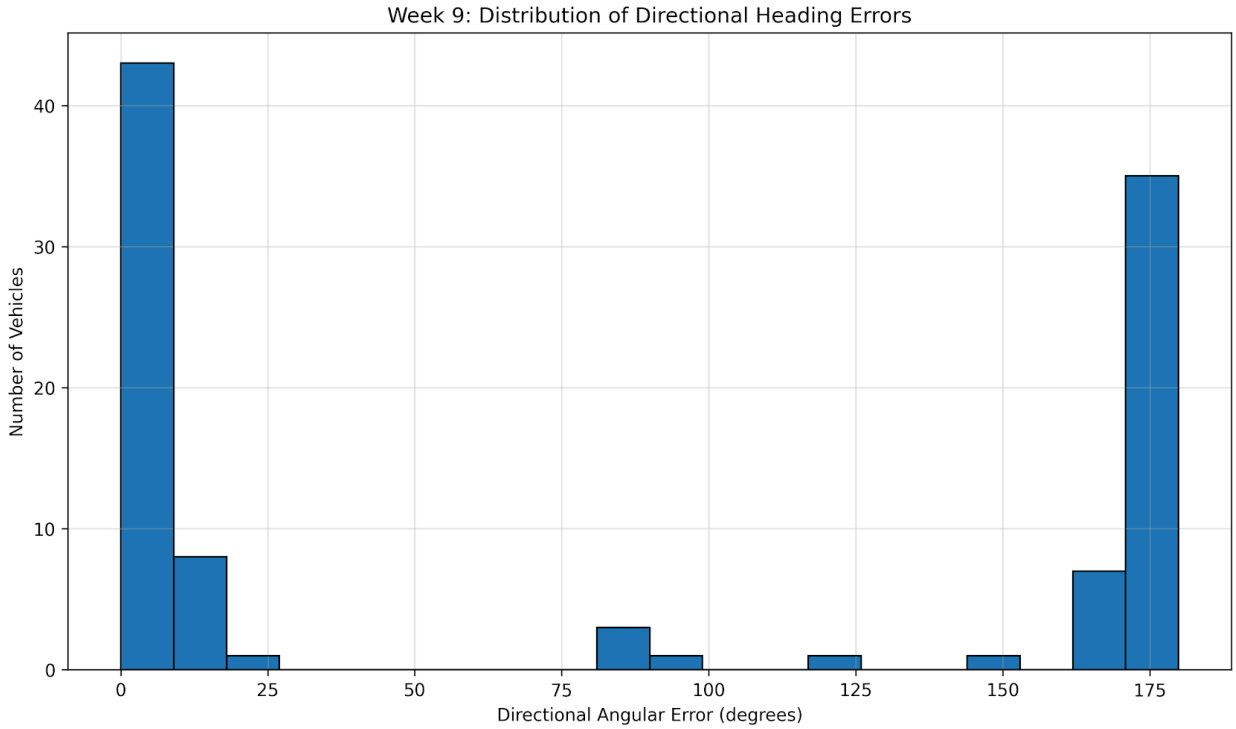


Figure 2. Histogram showing the distribution of directional heading errors across the 100 vehicle observations.

3.4 Class-wise Comparative Analysis

The performance of the four orientation estimation methods was also investigated for 11 different vehicle types.

The class-wise analysis enabled the identification of the differences in estimation errors for the various vehicle shapes and categories.

The Modified Hough results are summarized below.

Vehicle Class	Number of Samples	Mean Error
Auto3WCargo	3	5.06°
AutoRicksaw	8	4.77°
Bus	2	0.75°
Container	3	0.59°
MotorCycle	30	6.13°
PickUp	4	2.05°
SUV	34	5.80°
Sedan	7	1.78°
Tanker	1	1.29°
Trailer	3	0.42°
Truck	5	14.91°

Table 4. Class-wise performance of the Modified Hough Transform

The results show that the Modified Hough method produced relatively low mean errors for several vehicle categories.

The lowest mean errors were observed for Trailer, Container, and Bus. However, these categories had only three, three, and two observations respectively.

The Truck category had the highest mean error of 14.91° among the Modified Hough results. The SUV category also had a maximum error of 84.94°, even though its mean error was 5.80°.

The grouped comparison of all four methods was also generated to examine their performance across the different categories.

Since the number of observations varied across vehicle classes, the class-wise results should be interpreted with this limitation in mind.

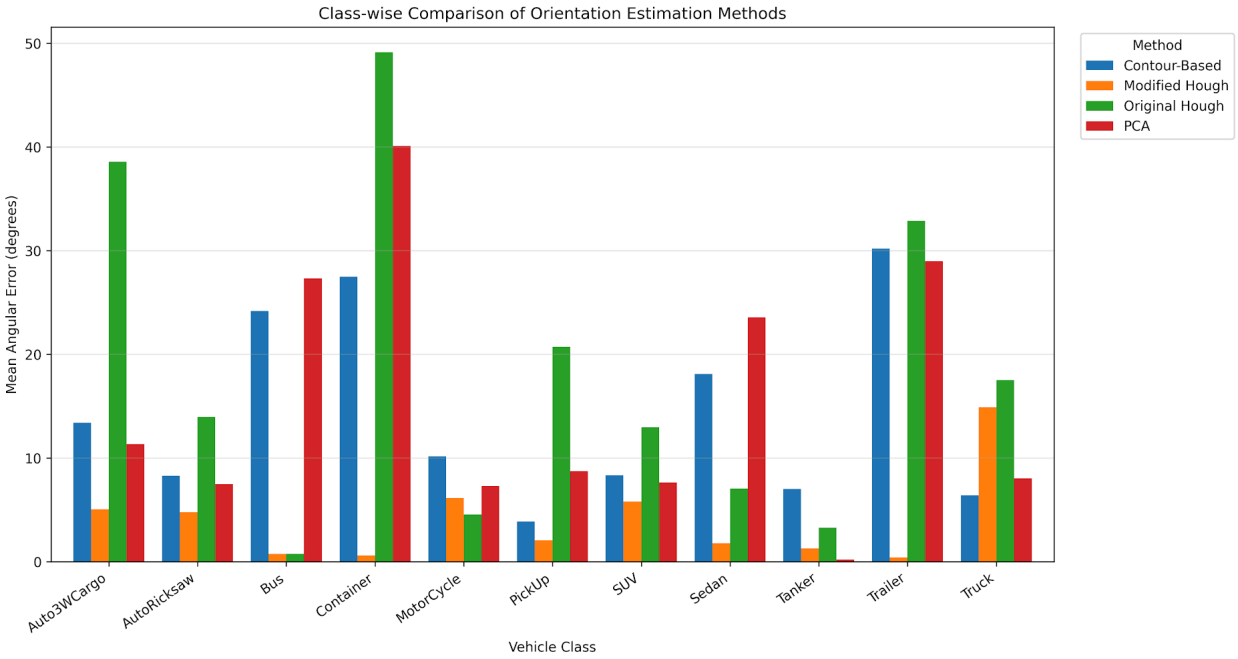


Figure 3. Class-wise comparison of angular errors for PCA, Original Hough, Contour-Based, and Modified Hough methods.

3.5 Analysis of the Highest Directional Heading Errors

The directional heading predictions were sorted according to their angular errors, and the 15 observations with the highest errors were saved separately.

The ten observations with the highest errors are presented below.

Vehicle ID	Vehicle Class	Ground-Truth Heading	Estimated Heading	Angular Error
OyGP_UjOQ_hJIP_vehicle_1	Bus	86.00°	266.09°	179.91°
ZKIx_Msfo_O7V9_vehicle_1	SUV	356.78°	177.02°	179.76°
WB91_YzXz_82r4_vehicle_1	Container	1.00°	181.36°	179.64°
y0bo_oHBY_y2mz_vehicle_1	Container	6.00°	185.51°	179.51°
Yfx3_zEFB_Ld6Y_vehicle_1	SUV	350.48°	171.25°	179.23°
zarI_WVuj_0DCc_vehicle_1	Container	6.00°	185.18°	179.18°
9SAd_tJKX_KWb9_vehicle_1	SUV	184.66°	3.18°	178.52°
Df3T_rTNG_KSCM_vehicle_1	MotorCycle	173.64°	351.82°	178.18°
ON14_goh0_iF0V_vehicle_1	SUV	183.74°	5.74°	178.00°
hwun_OlDq_ey5G_vehicle_1	Auto3WCargo	270.00°	92.02°	177.98°

Table 5. Top 10 vehicle observations with the highest directional heading errors

Several of these observations had angular errors close to 180°, indicating that the estimated heading was nearly opposite to the ground-truth direction.

This suggests that the front–rear selection stage was not always able to distinguish the correct direction along the estimated vehicle axis.

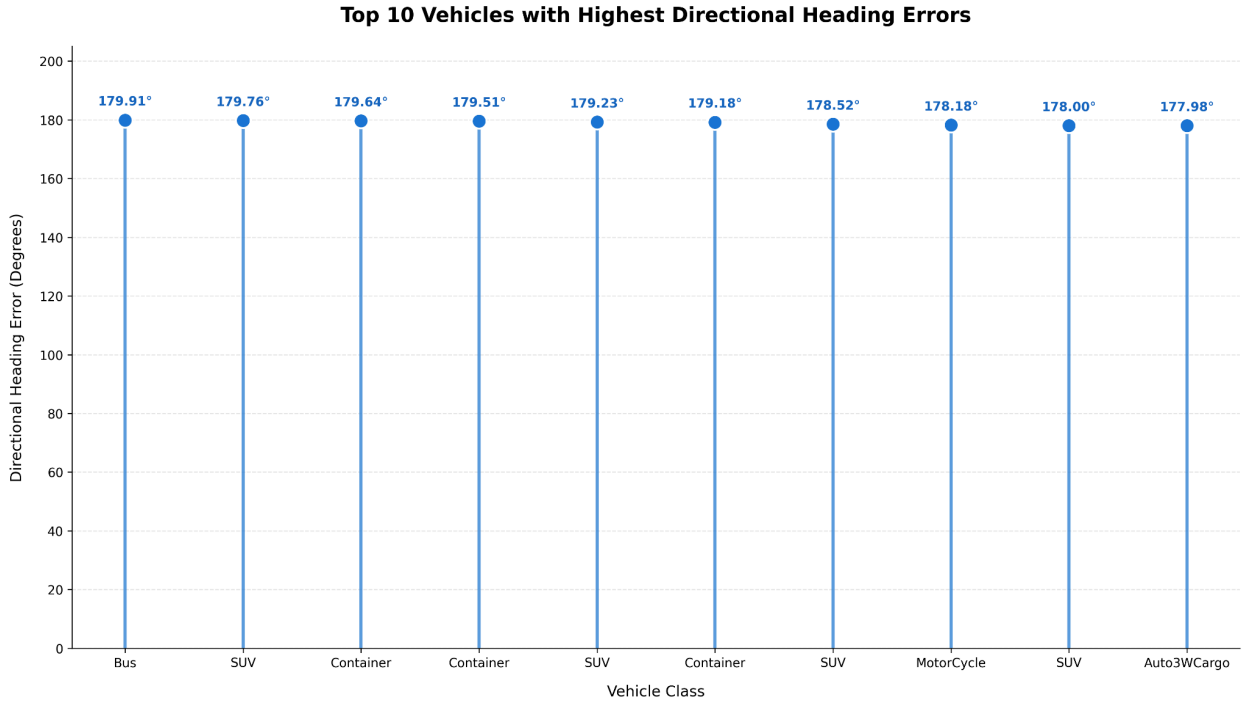


Figure 4. Ten vehicle observations with the highest directional heading errors.

Among these ten observations, the vehicle class distribution was:

Vehicle Class	Number of Observations
SUV	4
Container	3
Bus	1
MotorCycle	1
Auto3WCargo	1

SUV and Container vehicles appeared most frequently among the ten highest-error observations. However, this represents only the selected high-error cases and does not establish that these categories have the highest overall directional error rates.

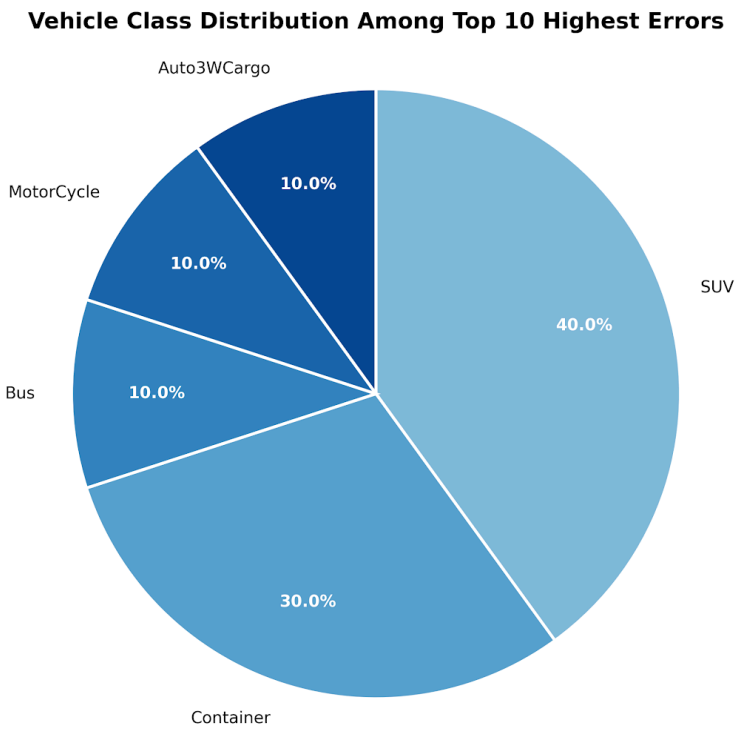


Figure 5. Vehicle class distribution among the ten observations with the highest directional heading errors.

3.6 Analysis of Failed Predictions

A separate analysis was performed to identify vehicle observations for which the directional heading estimation pipeline did not generate a valid prediction.

The final results showed that:

- Total vehicle observations: 100.
- Valid directional predictions: 100.
- Failed directional predictions: 0.
- Prediction coverage: 100%.

No failed directional predictions were found in the final dataset.

However, the angular error analysis showed that generating a prediction does not necessarily indicate that the predicted direction is accurate.

For comparison, the Original Hough method had two unavailable predictions in the undirected orientation evaluation, while PCA, Contour-Based, and Modified Hough generated predictions for all 100 observations.

The directional failed prediction records were saved separately for completeness.

3.7 Ground-Truth and Estimated Directional Heading Comparison

A scatter plot was generated to compare the ground-truth directional headings with the corresponding estimated headings.

The horizontal axis represents the ground-truth heading, while the vertical axis represents the estimated heading.

A reference diagonal was included to represent ideal agreement between the two values.

The closer the observations are to the reference diagonal, the more similar the estimated and actual headings are, and the farther away the observations are from the reference diagonal, the more different the headings are.

The plot displays a visual representation of the direction estimates over the entire range of 0° to 359°.

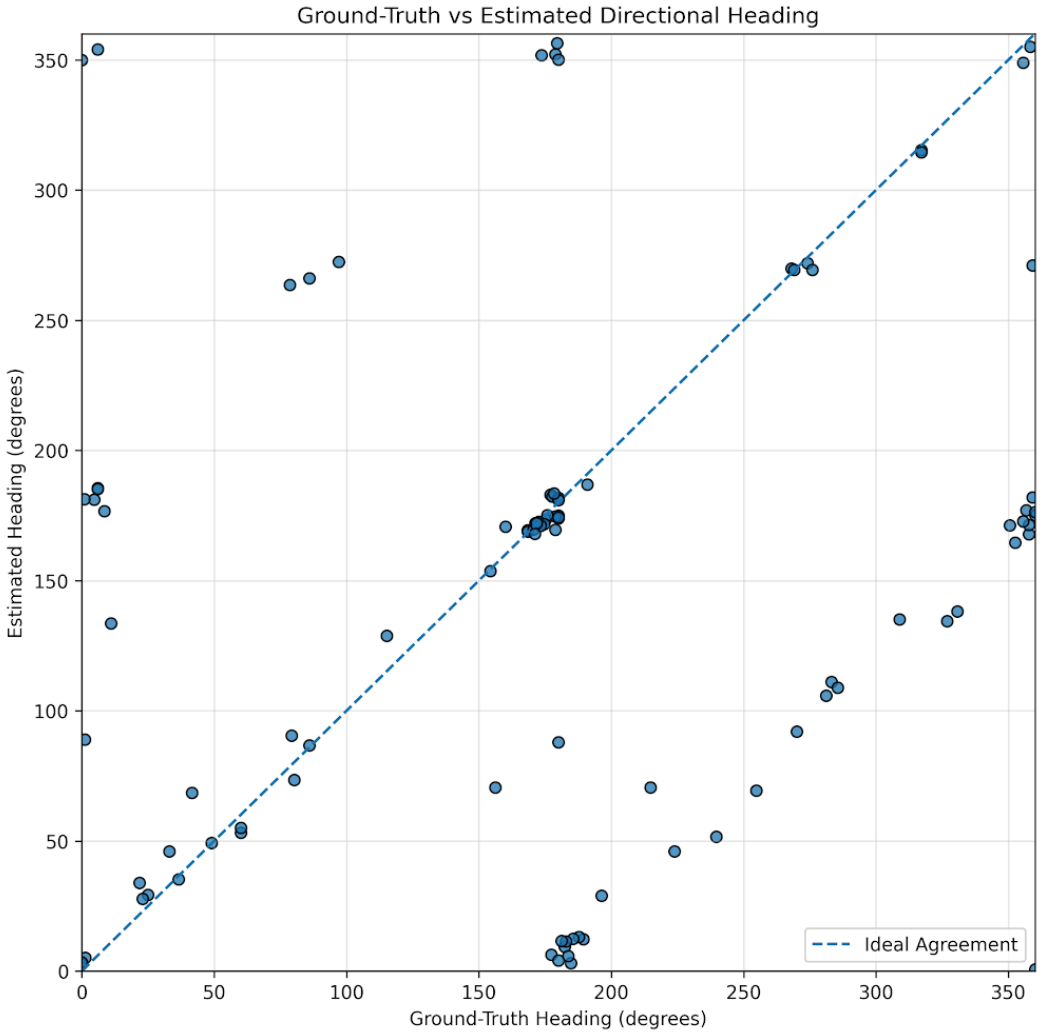


Figure 6. Comparison of ground-truth and estimated directional headings for the 100 vehicle observations.

3.8 Visual Inspection of Directional Heading Predictions

Visual comparisons were generated for all 100 vehicle observations to examine the estimated directional headings on the original vehicle crops.

For each vehicle, two arrows were displayed:

- Green arrow: Estimated directional heading.
- Red arrow: Ground-truth directional heading.

The vehicle ID, ground-truth heading, estimated heading, and corresponding angular error were also displayed.

The visualizations helped identify observations where the estimated and ground-truth arrows were approximately aligned and cases where the estimated direction differed substantially.

In particular, the visual inspection provided a way to examine the predictions with large angular errors and observe instances where the arrows pointed in nearly opposite directions.

To make the visualizations easier to inspect and avoid generating one excessively large output, the 100 observations were divided into five batches of 20 vehicles each.

4. Challenges Faced

During the final evaluation, we encountered and examined the following challenges:

1. Difference Between Undirected Orientation and Directional Heading

Estimating the longitudinal axis of a vehicle and identifying its actual front-facing direction are two different tasks. Although the Modified Hough method produced relatively accurate undirected orientation estimates, resolving the 180° ambiguity remained difficult.

2. Silhouette-based Front–Rear Identification

The silhouette asymmetry method is based on the assumption that the distribution of foreground pixels can be used to distinguish the front and rear of a vehicle. This is not true for all vehicle shapes, however, and thus the direction of the observations is not correct in some cases.

3. Background Interference During Silhouette Extraction

Otsu thresholding may contain unwanted areas like shadows, road markings, and other background elements. These areas may impact the silhouette mask and comparison between the two halves.

4. Vehicle Shapes Variation

There are various types of vehicle body structures. In some instances, the silhouettes are almost symmetrical and it is not easy to determine the direction from the foreground pixel distribution alone.

5. Large Directional Heading Errors

The final directionality evaluation revealed that some estimated headings were within about 180° of the actual direction. The mean angular error was significantly increased in these cases, although the median error was relatively low.

6. Differences between vehicle types

The analysis by class revealed that the methods were not equally effective in all vehicle classes. There were also limited observations in some categories, which made it challenging to draw firm conclusions from the results of individual observations.

7. Working with circular angular measurements

The direction heading evaluation needed to be done in circular angular error and not just in absolute angular difference. This was required to accurately assess the headings near the 0° and 360° limits.

These challenges enabled us to gain insight into the drawbacks of traditional image processing techniques and the distinction between the estimation of a vehicle's axis and the estimation of the full vehicle directional heading.

5. Final Project Status and Completed Deliverables

The final evaluation, analysis and documentation tasks were finished at the end of Week 9.

The final outputs were arranged in the Week9_Results folder in separate folders for tables, graphs, and visualizations.

6. Overall Project Conclusion

The project aimed to estimate vehicle orientation and directional heading using conventional digital image-processing techniques on the DRASHTI-HaOBB dataset.

The final evaluation brought together the results of the four undirected orientation estimation methods: PCA, Original Hough Transform, Contour-Based Estimation, and Modified Hough Transform.

Among these methods, Modified Hough achieved the best overall performance, with a Mean Absolute Error of 5.35° and a median angular error of 2.02° on the selected 100 vehicle observations.

The project was further extended to estimate the complete directional heading of vehicles in the range of 0°–359° using Modified Hough-based axis estimation and silhouette asymmetry.

Although the directional method generated predictions for all 100 vehicle observations, its Mean Absolute Error was 82.02°. The large difference between the mean and median errors, along with the presence of several near-opposite directional predictions, showed that the silhouette-based front–rear selection method was not consistently reliable.

The final results show that the conventional image-processing methods can be effective for the undirected vehicle orientation estimation from the selected observations. Estimating the full directional heading is more difficult, however, especially if it is based solely on silhouette shape and asymmetry.

In general, the project was able to implement and compare several traditional image processing methods, analyze the results numerically and visually, and record the limitations encountered in the experiments.

With the final evaluation, comparative analysis, visualizations, and result files completed, the project work has been brought to completion.

7. References

[1] Y. Bhavsar, M. Zaveri, M. Raval, K. R. Patel, and S. B. Zaveri, "Descriptor: Drone Nadir-view Annotated Images of Vehicles Detection Dataset for India with Heading-angle Oriented Bounding Box (DRASHTI-HaOBB)," *IEEE Data Descriptions*, 2026. Available at: https://ieeexplore.ieee.org/document/11421776.

[2] Y. Bhavsar, M. Zaveri, M. Raval, S. Zaveri, and Ahmedabad University, "DRASHTI-HaOBB: Drone nadir-view annotated images of vehicles dataset for India - heading-angle oriented bounding box," *Zenodo*, 2026. Available at: https://zenodo.org/records/18278989.